

Research Methods and Professional Practice

Literature Review Assignment

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1. Introduction

1.1 Aim and Research Focus

The construction sector currently faces pressure to modernize its ways. Projects are getting larger and more complex. Need coordination. The construction sector still relies on software systems that were built as rigid applications. Software systems are hard to update or scale when project conditions change. Microservices Architecture offers another way. Microservices Architecture divides software into components.

Exploring how MSA is used in web-based construction tools. If MSA can really support the industry's growing need for scalable and quickly adaptable systems. BY also compares MSA with software designs to determine whether MSA's advantages apply equally to medium and large construction companies.

1.2 Intended Readership

For readers who care about digitalization in computing and construction. The review is for software engineers, construction informatics researchers, strategy consultants, and managers who select or develop IT systems. The intention is to provide researchers with a consolidated understanding of the current literature and to give practitioners an idea of whether microservices can improve construction workflows.

1.3 Rationale and Importance

The construction projects increasingly rely on digital tools. The digital tools are BIM platforms, sensors, AI-based prediction systems, and mobile coordination apps (Li et al., 2021). The digital tools let the team manage the resources and track the progress. The digital tools produce a lot of data. The older systems cannot handle the data well.

MSA catches the eye because MSA lets software grow step, by step. MSA does not impose a specific system shape. MSA also helps teams roll out updates reliably (Newman, 2015; Dragoni et al, 2017). Construction faces its problems. Construction teams deal with design changes. Construction crews work on the move. Construction sites often have connectivity. Understanding how microservices solve these problems matters to organizations that want to modernize their infrastructure. From my view, microservices give construction teams a way to adapt quickly and keep their systems stable.

1.4 Approach to Literature Selection

This review draws on secondary sources collected through searches on IEEE Xplore, ScienceDirect, ACM Digital Library, and Google Scholar. I used the keywords microservices architecture, construction IT systems, BIM platforms, DevOps, in construction, and cloud-native applications to find the sources. The review focuses on publications from 2014–2025, ensuring that both foundational theories and emerging case studies are represented. Sources were included if they discussed

MSA directly or analyzed software environments that face similar coordination and scalability challenges to construction. Each source was evaluated for relevance, clarity, and contribution to current discussions on digital construction systems.

2. Conceptual Foundations

2.1 Understanding Microservices Architecture (MSA)

Microservices Architecture changes the way we build software. Microservices Architecture builds each feature—scheduling, authentication, document handling — as a service, with its own logic and data. Microservices Architecture services communicate with each other via methods, such as REST APIs or event-driven messages. Those lightweight methods let each service run without relying on the services (Newman, 2015).

Bounded contexts mean that each service owns its domain model and database. The separation gives the system better fault isolation. When one service fails, the rest of the system continues to run. The design also lets teams update or redeploy a component without touching the platform. The MSA's autonomy helps change environments. The system must adapt quickly to requirements. (Fowler, 2014)

2.2 Transition from Monolithic Systems to Modular Designs

The monolithic applications are easy to build at first. The monolithic applications become hard to maintain as more features are added. Updating one part often forces the system to be rebuilt and deployed. That adds delays. It can cause instability.

Service-Oriented Architecture (SOA) tried to split applications into pieces. The Service-Oriented Architecture (SOA) implementations often need middleware and complex orchestration tools. Microservices go a step further. Microservices stress decentralization. Microservices use units that can be deployed independently. Microservices rely on container technologies such as Docker and Kubernetes (Pahl et al. 2018).

The construction workflows face project conditions that often change. The modularity is very useful in those situations. From observation, when a contractor needs to add a compliance-checking function to a project, the MSA allows the teams to add the new service without stopping other operations. Adaptability helps teams keep improving and reduces downtime. The downtime matters because the project delays bring schedule problems.

3. Construction Industry's Technological Demands

3.1 Ongoing Digital Transformation in Construction

Construction has moved slowly in using tools for years. The construction industry now feels a push toward modernization. Rising project complexity, tighter regulations, labor shortages, and higher expectations for transparency expose the limits of manual and paper-based workflows. Recent studies show that companies that invest in technologies such as BIM, digital twins, on-site sensors, and AI-driven analytics achieve gains in project coordination and cost efficiency. Digital technologies

generate data sets. Companies must. Process the varied data sets fast so that real-time decisions can be made (Oesterreich & Teuteberg, 2016).

Connecting the technologies into a single system is challenging. In the construction industry, many platforms operate in isolation. Those platforms create data silos. A flexible software architecture, such as MSA, can unify technologies. MSA allows new digital components to be added without rebuilding the system from scratch.

3.2 Core Requirements for Construction Web Applications

Construction projects involve stakeholders with diverse responsibilities. The stakeholders are architects, engineers, site supervisors, and subcontractors. The stakeholders often work in places. Because of those digital tools, construction must support collaboration. The digital tools for construction must work on devices. Digital tools for construction must also safeguard data.

Key requirements include:

Relying on real-time updates so the team member can view the project information. Real-time updates let the team member see the project information as soon as it changes. Real-time updates keep the team member aware of the project information. Real-time updates work for the team member. Cross-platform accessibility is needed on the site. Cross-platform accessibility works best when tablets and mobile phones are used a lot. Integration with existing systems, such as BIM platforms, scheduling tools, and financial software. Offline capability, important for remote sites with unstable internet connections. Role-based access control limits who can see the project data. Role-based access control ensures that only authorized users can access project data.

Microservices architecture matches the needs. A permissions service can manage user roles on its own. A scheduling service can handle schedules. A document storage service can keep files. An equipment tracking service can track tools. When a service has a fault, the fault does not bring down the rest of the application. That matters in construction because downtime can stop work.

Also, MSA lets each service scale independently. MSA enables tasks such as analytics and 3D rendering leverage additional processing power. MSA does not slow down functions. The adaptability of MSA matches the changing demands of construction projects.

4. Microservices Applications in Construction

4.1 Practical Implementations in Industry Platforms

The real test of microservices lies in how effectively they function in live construction environments. Several well-known industry platforms have already adopted microservices-based designs, providing useful examples of how this architecture performs in practice.

Autodesk Construction Cloud uses containers to handle tasks such as document coordination, version control, and project scheduling. Autodesk Construction Cloud updates each service on its own, so the global teams can see the information and

keep working without stopping. I have used Autodesk Construction Cloud. I notice the updates do not interrupt the work. Trimble Connect splits functions, such as design collaboration, spatial coordination, and model visualization, into microservices. Trimble Connect lets users turn on the features they need. Additionally, Procore's platform uses an event-driven approach. Procore's platform pushes real-time alerts to the site teams when schedules shift, when safety incidents happen, or when important updates are uploaded. The communication powered by microservices helps stop delays and makes on-site awareness better. Smaller companies try microservices. One example is a cloud-based procurement system that automates purchase orders, delivery tracking, and vendor compliance. The microservices, with built-in logging and API integrations, let the system cut processing time and improve transparency and accuracy.

4.2 Functions Tailored to Construction Workflows

Microservices offer particular benefits when developing tools that address construction-specific tasks. For example, a safety monitoring system may rely on separate microservices to collect sensor data, analyze risk indicators, issue alerts, and generate reports. If a new type of wearable device is introduced on-site, only the data-collection service needs to be modified, leaving the rest of the system untouched. Analytical tools also benefit from microservices. Services dedicated to machine-learning predictions or cost-forecasting models can scale up processing resources when required, without increasing the load on basic features such as file storage or task lists. This independence between services helps keep system performance stable even during peak periods. It also supports cost-efficient cloud usage because only high-demand services consume additional computing power. For construction firms dealing with fluctuating workloads across different project phases, this flexibility is especially valuable.

5. Analytical Framework

5.1 Key Dimensions for Evaluation

To assess how well microservices architecture helps construction-focused web applications, the literature often highlights recurring evaluation criteria. The literature lists a few evaluation criteria for microservices architecture in construction-focused web applications.

Scalability is essential because construction projects can have hundreds of users using the system simultaneously, especially during peak coordination periods. Scalability keeps the system running under high demand. Microservices let each component—such as analytics engines or document storage—scale up on its own as demand increases. Microservices let the analytics engines or the document storage grow without stopping the rest of the system. Maintainability means how easy the system is to update, repair, or change. Each microservice operates as a unit, allowing developers to apply patches. Add new features without rebuilding the whole application. Deployment speed is essential in a sector where unexpected changes, regulatory updates, or safety requirements can appear during a project. Deployment speed lets teams respond quickly. Microservices support integration and continuous delivery pipelines. Microservices enable rollouts.

Team autonomy helps us innovate and stay productive. Team autonomy lets us try ideas and finish work faster. With microservices, development teams can each work on a service in parallel. Development teams can do that without stepping

on each other's progress. Lastly, system resilience—the ability to keep the system running when a component fails—is essential in construction. In a construction system, resilience matters because software downtime can stop work on the site and delay the project schedule. This happens often.

5.2 Alignment with Agile and DevOps Practices

Microservices architecture fits well with development methods and development workflows. Agile development methods and DevOps workflows both focus on making changes step by step and on deployment. Docker lets developers package services into containers. Kubernetes moves services. Scales services. CI/CD tools, such as Jenkins, for testing and updates. Jenkins helps teams deliver changes often and with risk. (Bass et al., 2015). Monitoring tools, such as Prometheus and Grafana, give developers a view of system performance. I rely on monitoring tools. Monitoring tools let developers spot problems early and roll back updates when needed. Monitoring tools are especially useful in construction because the construction software must change quickly as the project needs shift.

Microservices and developers together form the foundation for building adaptable IT systems. Microservices and DevOps let the system grow and change without breaking.

6. Findings

6.1 Documented Advantages in the Literature

Both academic and industry sources say that microservices architecture works well for software systems that handle large volumes of data and require frequent updates. Those conditions are common in construction projects. The microservices architecture lets a software system be extended or changed without breaking parts. For example, a construction firm can add a drone data processing module or a new compliance checker without rebuilding the application. Improved integration is another advantage. Construction projects depend on platforms. The platforms are BIM tools, scheduling apps, cost-management systems, and on-site sensors. Microservices talk through APIs. The lightweight APIs make it easy for platforms to connect. The easy connection keeps data exchange smooth. Improved integration helps construction projects run smoothly.

Multi-tenancy is a point that the literature mentions. A single microservices-based platform can serve contractors or project teams at the same time. The single microservices-based platform gives each contractor or project team permissions and different data structures. This helps firms that manage projects at once. The firms do not need to duplicate the system for each client. The microservices have potential. Microservices can improve the flexibility, performance, and long-term sustainability of construction software ecosystems.

6.2 Commonly Reported Limitations

Despite the clear benefits, microservices also introduce challenges that must be managed carefully. Many studies point out that the initial setup of a microservices system is often more expensive and complex than building a simple monolithic

application (Taibi et al., 2020). Firms need developers with specialized knowledge of distributed systems, container orchestration, and DevOps pipelines—skills that are not always readily available in the construction sector. Security becomes more demanding as well. Instead of protecting a single central application, organizations must secure communication among numerous small services and ensure data remains consistent across different storage units. Troubleshooting can be harder, too. When a fault occurs, it may involve interactions across several services, making it more time-consuming to identify the exact cause. Smaller firms with limited IT resources may find these requirements overwhelming unless they adopt microservices gradually or seek external technical support. These limitations suggest that while microservices offer major advantages, their success in construction settings depends heavily on organizational readiness, infrastructure maturity, and access to skilled technical teams.

7. Strengths and Limitations of the Literature

7.1 Strengths of Existing Research

The literature on microservices architecture, also called MSA, gives a base for understanding how MSA works in systems. The literature on microservices architecture explains the principles. The architectural principles include coupling, independent deployment, and bounded contexts. The literature on microservices architecture illustrates the principles with examples from many industries. Construction-specific research is still new. Several papers compare construction IT systems with IT systems in finance, healthcare, and logistics. Cross-industry insights strengthen the argument that microservices can handle workflows, real-time coordination, and multi-stakeholder environments.

The early case studies, from construction technology companies that show results. Early case studies highlight improvements in modularity, integration, and system reliability, and suggest that MSA aligns well with construction firms' change goals.

7.2 Weaknesses and Gaps in the Literature

Despite these strengths, the current literature also has notable limitations. Much of the existing evidence relies on general software engineering research rather than construction-specific implementations. Because of this, certain practical issues—such as limited connectivity on job sites, offline work requirements, and the need to synchronize field devices—are not explored in detail. Another gap is the lack of long-term empirical studies. Most available research focuses on prototypes, pilot systems, or short-term evaluations. This means there is limited understanding of how microservices perform across multiple construction project phases or within organizations that operate many projects simultaneously.

The literature also lacks a consistent set of criteria for evaluating the adoption of microservices in construction. While some studies emphasize scalability and deployment speed, others highlight cost, security, or organizational readiness. Without a common framework, it becomes difficult to compare findings or to determine which factors matter most to construction firms considering migration to an MSA.

Overall, the weaknesses suggest a need for more targeted, empirical research that reflects the real-world challenges and operational rhythms of construction work.

8. Discrepancies in the Literature

The academic and industry discussions about microservices architecture (MSA) are lively. The academic and industry discussions show places where opinions differ. The academic and industry discussions also show that implementing a microservices architecture (MSA) on real-world construction sites is challenging.

One common point of disagreement is the complexity that microservices bring. Some writers argue that breaking an application into services provides the system with greater operational clarity, better scalability, and easier maintenance over time. Other writers say that microservices can become a burden when there are microservices that make deployments harder to coordinate, and that microservices increase the amount of communication between services. Cost-efficiency gives results. Larger organizations with established IT teams see long-term gains when they adopt microservices. The larger organizations report the gains. The smaller firms, the firms without developed expertise, often face high initial costs for infrastructure, training, and system redesign. The smaller firms struggle with the cost. The literature shows outcomes. The mixed outcomes depend on the size and the technological maturity.

Researchers debate the size of a microservice. In my experience, some recommend focused microservices to maximize independence. Others warn that making a microservice granular increases network traffic, slows performance, and makes debugging more difficult (Balalaie et al., 2016). These differences show that microservices are not a one-size-fits-all solution. In my experience, a successful microservices implementation depends on the project size, the available expertise, the cloud infrastructure, and the specific demands of construction workflows. The variety of opinions shows that a firm must shape its microservices strategy for its working environment rather than relying solely on general recommendations in the literature.

9. Conclusion and Recommendations

9.1 Overall Conclusions

The literature indicates that microservices architecture offers a strong and adaptable foundation for developing modern construction web applications. Its modular design, independent deployment capabilities, and alignment with cloud-native development make it well-suited to the fast-paced and often unpredictable nature of construction projects.

Case examples from industry platforms demonstrate that MSA can improve system responsiveness, enhance collaboration, simplify integration with external tools, and support continuous updates without major disruptions. These benefits are especially relevant as construction firms increasingly rely on BIM data, IoT devices, and advanced analytics.

However, the successful adoption of microservices depends heavily on organizational preparedness. Firms must have access to reliable infrastructure, stable connectivity, and teams capable of managing distributed systems. Smaller companies or those with limited technical expertise may encounter challenges related to cost, complexity, and maintenance requirements.

In summary, while microservices show strong potential to modernize construction IT systems, their effectiveness depends on how well the architecture aligns with the firm's digital maturity, project needs, and long-term technology strategy.

9.2 Recommendations for Practice and Future Research

For construction companies considering adopting microservices, a gradual, well-planned approach is advisable. Beginning with a small, low-risk module—such as document storage, scheduling, or user management—can help teams familiarize themselves with the architecture while avoiding major disruptions to ongoing projects. Investing in DevOps tools, cloud infrastructure, and staff training will also increase the likelihood of a successful transition. Combining microservices with legacy systems early on can reduce operational risk and enable firms to modernize incrementally. Collaboration between software vendors, construction managers, and IT teams is essential for developing industry-specific best practices and ensuring that new systems meet the practical needs of on-site workflows. From an academic perspective, more empirical research is required. Long-term studies evaluating microservices across multiple project phases, investigations into offline-capable service design for remote sites, and comparative analyses between microservices and hybrid architectural models would all contribute valuable insights. Overall, a carefully managed adoption strategy—supported by appropriate tools, skills, and research—can help construction organizations harness the full potential of microservices while minimizing associated challenges.

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